BetterAuth Implementation Backlog

Pre-requisites:

- A valid Amazon AWS account with access to AWS Cognito
- Windows 10 or later or Windows Server 2016 or later
- An Intel Pentium 4 processor or later that is SSE3 capable

The BetterAuth implementation begun with AWS Cognito being setup. This included adding Federation Identity Providers and setting up the callback URLs, sign-in URLs and sign-out URLs. Developer accounts needed to be made for Google and Amazon to get client IDs and client secrets to add them as Federation Identity Providers, and the Amplify deployment URLs of the web app were included for the URLs.

Items selection	User pool name	User pool ID	Created time	Last updated time
	User pool - ezsa3	ap-southeast-2_Do0ZVzLo7	16 hours ago	16 hours ago

App client information

Edit

App client name

Clockwork-MVP-Web-App

Client ID

un0t3utbt8p4a77jfang6fe3p

Client secret

-

Authentication flows

Choice-based sign-in

Secure remote password (SRP)

Get user tokens from existing authenticated sessions

Authentication flow session duration

3 minutes

Refresh token expiration

5 day(s)

Access token expiration

60 minutes

ID token expiration

60 minutes

Advanced authentication settings

Enable token revocation

Enable prevent user existence errors

Created time

May 19, 2025 at 21:49 GMT+10

Last updated time

May 20, 2025 at 10:27 GMT+10

Managed login pages configuration

Info

Edit

View login page

Configure the managed login pages for this app client.

Status

Available

Allowed callback URLs

https://lab.d3gs10k0enapyd.amplifyapp.com/

Allowed sign-out URLs

https://lab.d3gs10k0enapyd.amplifyapp.com/login

Identity providers

Google

LoginWithAmazon

OAuth grant types

Authorization code grant

OpenID Connect scopes

email

openid

phone

profile

Custom scopes

-

Items selection	Group name	Description	Precedence	Created time
	ap-southeast-2_Do0ZVzLo7_Google	Autogenerated group for users who sign in using Google	-	16 hours ago
	ap-southeast-2_Do0ZVzLo7_LoginWithAmazon	Autogenerated group for users who sign in using LoginWithAmazon	-	13 hours ago

Following the setup of the AWS services, Amplify libraries needed to be installed using npm to allow the BetterAuth authentication to occur through the setup Cognito user groups.

npm install aws-amplify

npm install @aws-amplify/auth

Furthermore, App.js was updated to call a configuration of Amplify when the website loads, and aws-exports.js were updated to use the correct callback and sign-out URLs.

App.js section of code:

import { BrowserRouter as Router, Route, Routes } from 'react-router-dom';

import "./App.css";

import Login from "./pages/Login/Login";

import Layout from "./components/Layout/Layout";

const App = () => {

Was updated to:

import { BrowserRouter as Router, Route, Routes } from 'react-router-dom';

```
import "./App.css";

import Login from "./pages/Login/Login";

import Layout from "./components/Layout/Layout";

import { Amplify } from 'aws-amplify';

import awsExports from './aws-exports';


Amplify.configure(awsExports);


const App = () => {
```

aws-exports.js section of code:

```
    "redirectSignIn": "http://localhost:3000/",

    "redirectSignOut": "http://localhost:3000/",
```

Was updated to:

```
    "redirectSignIn": "https://lab.d3gs10k0enapyd.amplifyapp.com/",

    "redirectSignOut": "https://lab.d3gs10k0enapyd.amplifyapp.com/login",
```

Many attempts were then made to try to get the web app running using Amplify deployments and their associated links. None of these attempts worked however, due to errors consistently occurring with ClockWorkWebLabbing/frontend/src/pages/Login/Login.tsx. Other attempts were due to the aws-exports.js file, however those were resolved.

Errors and their associated update to the code are listed below:

Code:

```
import { signInWithRedirect } from "aws-amplify/auth";
```

Was updated to:

```
import { Auth } from 'aws-amplify';
```

Produced Error:

```
178 2025-05-20T01:44:39.853Z [INFO]: Failed to compile.

179 2025-05-20T01:44:39.858Z [INFO]: Attempted import error: 'Auth' is not
exported from 'aws-amplify' (imported as 'Auth').

180 2025-05-20T01:44:39.976Z [ERROR]: !!! Build failed

181 2025-05-20T01:44:39.977Z [ERROR]: !!! Error: Command failed with exit code 1
```

Code:

```
import { Auth } from "aws-amplify";

...

await Auth.federatedSignIn({ provider });
```

Was updated to:

```
import Amplify, { Auth } from "aws-amplify";

...

await Amplify.Auth.federatedSignIn({ provider });
```

Produced Error:

```
179 2025-05-20T01:59:25.764Z [INFO]: Attempted import error: 'Auth' is not
exported from 'aws-amplify' (imported as 'Amplify').

180 2025-05-20T01:59:25.880Z [ERROR]: !!! Build failed
```

181 2025-05-20T01:59:25.880Z [ERROR]: !!! Error: Command failed with exit code 1

Following the run in with these errors, a different approach was taken to reinitialize the Amplify folder by virtually deleting and reinstalling it. However, this was met with further errors.

A large portion of these errors were to do with an inability to access the correct files and authorize myself as the correct user, and as a result are not documented. However, other errors are seen below that were unexpectedly slowing progress. These errors have made me realize that majority of the Amplify files are mismatched or at the very least my local files, the github files, and Angus' computer's files all having files to do with this implementation is making it more difficult. This is seen with attempting to find and make changes to the right files proving much more difficult than what would be expected.

C:\Users\James\ClockWorkWebLabbing\frontend\amplify>amplify auth update

🛑 File at path:

'C:\Users\James\ClockWorkWebLabbing\frontend\amplify\backend\amplify-meta.json' does not exist

C:\Users\James\ClockWorkWebLabbing>amplify status

🛑 No Amplify backend project files detected within this folder.

C:\Users\James\ClockWorkWebLabbing>amplify status

🛑 No Amplify backend project files detected within this folder.

? Please choose the profile you want to use Code-CrusadersAWS

🛑 App d3gs10k0enapyd not found.

Amplify AppID found: d3gs10k0enapyd. Amplify App name is: ClockWorkWebLabbing

🛑 Cannot find backend environment dev in Amplify Console app: ClockWorkWebLabbing

BackendEnvironment dev does not exist.

Amplify AppID found: d3gs10k0enapyd. Amplify App name is: ClockWorkWebLabbing

🛑 Cannot find backend environment Lab in Amplify Console app: ClockWorkWebLabbing

BackendEnvironment Lab does not exist.

Amplify AppID found: d3gs10k0enapyd. Amplify App name is: ClockWorkWebLabbing

🛑 Cannot find backend environment lab in Amplify Console app: ClockWorkWebLabbing

BackendEnvironment lab does not exist.

C:\Users\James\ClockWorkWebLabbing>amplify env add

🛑 No Amplify backend project files detected within this folder.

C:\Users\James\ClockWorkWebLabbing>amplify pull --appId d3gs10k0enapyd --envName lab

🛑 Amplify appId mismatch.

C:\Users\James\ClockWorkWebLabbing>amplify pull --appId d3gs10k0enapyd --envName lab

🛑 Amplify appId mismatch.

Following these errors, a different approach was taken. This initially included going back to the Login.tsx code and fixing any errors, and going methodically through every config file to find any mismatches in the data in those files and what was shown on AWS. The first update to the Login.tsx code after this new direction was taken was removing the usage of useAuthCheck, and opting to use signInWithRedirect only, with the providers specifically added in the async parameters.

Old code:

```
import React from "react";

import { Navigate } from "react-router-dom";

import { signInWithRedirect } from "aws-amplify/auth";

import useAuthCheck from "../../utils/useAuthCheck";

import Logo from "./clockwork-logo.png";

import "./Login.scss";


enum Provider {

  GOOGLE = "Google",

  AMAZON = "Amazon",

}


const Login: React.FC = () => {
```

```
 const isAuthenticated = useAuthCheck(); // Use the hook


 const handleSignIn = async (provider: Provider): Promise<void> => {
  try {
   await signInWithRedirect({ provider });
  } catch (error) {
   console.error("Error during federated sign-in:", error);
  }
 };


 if (isAuthenticated === null) {
  return <div>Loading...</div>; // Loading state
 }


 if (isAuthenticated) {
  return <Navigate to="/" replace />; // Redirect if already authenticated
 }
```

Updated code:

```
import React from "react";
import { signInWithRedirect } from "aws-amplify/auth";
```

```
const Login = () => {
  const handleFederatedLogin = async (provider: "Google" | "LoginWithAmazon") => {
    try {
      await signInWithRedirect({
        provider, // Must match Cognito identity provider names exactly
      });
    } catch (error) {
      console.error("Federated sign-in error:", error);
    }
  };
```

The next change to the code was to the try function, where it was found that “LoginWithAmazon” is not a valid AuthProvider, and as a result has to be implemented as a custom federated provider.

Updated code:

```
import React from "react";
import { signInWithRedirect } from "aws-amplify/auth";

const Login = () => {
  const handleFederatedLogin = async (provider: "Google" | "LoginWithAmazon") => {
    try {
      await signInWithRedirect({
```

```
      provider: provider === "Google" ? "Google" : { custom: "LoginWithAmazon" },
    });
  } catch (error) {
    console.error("Federated sign-in error:", error);
  }
};
```

Then it the code finally compiled, however the login button redirections did not work because the deployed origin did not match with the Oauth Redirect URIs. To fix this the following URIs were configured:

Allowed callback URLs

http://localhost:3000/

https://lab.crusaders.clockwork.vip

https://lab.crusaders.clockwork.vip/login

https://lab.crusaders.clockwork.vip/oauth2/idpresponse

Allowed sign-out URLs

http://localhost:3000/

https://lab.crusaders.clockwork.vip

Then it was found that there was a Cognito domain name mismatch so it was changed from:

https://clockworkae96e028-ae96e028-dev.auth.ap-southeast-2.amazoncognito.com

To:

https://ap-southeast-2do0zvzlo7.auth.ap-southeast-2.amazoncognito.com

Then it was found that there was a App Client ID mismatch in the config files so across all the config files it was changed from:

7eoiupdr6l5mt965lgpmuu01ac

To:

un0t3utbt8p4a77jfang6fe3p

Then the scopes were found to be causing errors since LoginWithAmazon could not handle some of them so they were changed in all the config files from:

```
"scope": [
        "phone",
        "email",
        "openid",
        "profile",
        "aws.cognito.signin.user.admin"
],
```

To:

```
    "scope": [
      "phone",
      "email",
      "profile"
    ],
```

Then it was found that Authorised redirect URIs were not added to the Client ID created in the Google Cloud Console for logging in with Google. The following were then added:

https://userpool-ezsa3.auth.ap-southeast-2.amazoncognito.com/oauth2/idpresponse

https://ap-southeast-2do0zvzlo7.auth.ap-southeast-2.amazoncognito.com/oauth2/idpresponse

https://lab.crusaders.clockwork.vip/

https://lab.crusaders.clockwork.vip/oauth2/idpresponse

Then it was found that the redirect URLs in the config files were again incorrect, being changed from:

"redirectSignIn": "https://lab.crusaders.clockwork.vip/",

"redirectSignOut": "https://lab.crusaders.clockwork.vip/",

To:

"redirectSignIn": "https://lab.crusaders.clockwork.vip/oauth2/idpresponse",

"redirectSignOut": "https://lab.crusaders.clockwork.vip/logout", // optional but cleaner

Then it was found that the Callback URLs for the App client settings were not updated with the new redirectSignIn link, and so it was changed accordingly.

Then it was found that the Allowed Return URLs and Allowed Origins for LoginWithAmazon were not added to the Client ID in the Amazon Developer Console, so they were added accordingly:

Allowed Origins	• https://lab.crusaders.clockwork.vip
Allowed Return URLs	• https://lab.crusaders.clockwork.vip/oauth2/idpresponse

- https://ap-southeast-2do0zvzlo7.auth.ap-southeast-2.amazoncognito.com/oauth2/idpresponse

Then it was found that the scopes for the LoginWithAmazon federated identity provider were not valid, so they were changed from:

email profile

To:

profile